

Full wreath-product monodromy for the square of an explicit Keller map

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with heavy assistance from ChatGPT 5.6 Sol

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Abstract

Let $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ be the newly announced noninjective polynomial map with constant Jacobian determinant -2 and geometric monodromy S_3 . We determine the geometric monodromy of its canonical self-composition:

$$\text{Mon}(F \circ F) = S_3 \wr S_3$$

in degree nine, of order 1296. On the target line $(1, 2, s)$, an exact degree-nine eliminant has a one-edge Newton polygon at infinity, giving a 9-cycle; its discriminant has a squarefree factor of multiplicity one, giving a single within-block transposition; and the outer cubic gives a transposition of the three blocks. An elementary group lemma forces the full wreath product. Postcomposition by $\text{diag}(1/4, 1, 1)$ gives a noninjective unit-Jacobian map realizing the pair $(9, S_3 \wr S_3)$. We also prove that every iterate $F^{\circ m}$ has a full 3^m -cycle in its geometric inertia at infinity. Exact SymPy, dependency-free finite-field, PARI/GP, and GAP checks accompany the argument.

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1 The map and the result

Put $u = 1 + xy$ and define $F = (F_1, F_2, F_3)$ by

$$\begin{aligned} F_1 &= u^3 z + y^2 u(4 + 3xy), \\ F_2 &= y + 3xu^2 z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 z. \end{aligned} \tag{1}$$

Exact expansion gives $\det JF = -2$. Moreover,

$$p_0 = (0, 0, -\tfrac{1}{4}), \quad p_1 = (1, -\tfrac{3}{2}, \tfrac{13}{2}), \quad p_2 = (-1, \tfrac{3}{2}, \tfrac{13}{2}) \tag{2}$$

all map to $(-1/4, 0, 0)$.

For a dominant polynomial map H , write $\text{Mon}(H)$ for the Galois group of the normal closure of the induced function-field extension, in its natural action on a generic fiber.

Theorem 1. *The self-composition $G = F \circ F$ has generic degree nine and*

$$\text{Mon}(G) = S_3 \wr S_3 \leq S_9, \quad |\text{Mon}(G)| = 6^3 \cdot 6 = 1296.$$

The action has three blocks of size three, indexed by the intermediate points in a generic fiber of the outer copy of F .

Proposition 2. *For every $m \geq 1$, the geometric monodromy of $F^{\circ m}$ contains a cycle of length 3^m .*

Proposition 3. *Let $\widehat{G} = (G_1/4, G_2, G_3)$. Then $\det J\widehat{G} = 1$, $\text{Mon}(\widehat{G}) = S_3 \wr S_3$, and $\widehat{G}$ is noninjective. The three points in (2) all map under $\widehat{G}$ to $(0, 0, -1/2)$.*

The last proposition follows immediately from the theorem and $\det J(F \circ F) = 4$.

2 The cubic inverse resolvent

For a target (a, b, c) , introduce

$$C_{a,b,c}(t) = 2at^3 - bt^2 + 2t - c. \quad (3)$$

For a simple root t , the generic preimage is reconstructed by

$$y = -\frac{bt^2 + 3c - 6t}{2t^2}, \quad x = \frac{t}{1 - ty}, \quad z = \frac{2x - 3x^2y - c}{x^3}. \quad (4)$$

Substitution reduces every coordinate of $F(x, y, z) - (a, b, c)$ to zero modulo $C_{a,b,c}(t)$. The parameter is $t = x/(1 + xy)$, reciprocal to the usual parameter $y + 1/x$.

The three roots of (3) parametrize the generic fiber. For $F \circ F$, each of three outer roots supports a second cubic of inner roots. Consequently the generic degree is nine and

$$\text{Mon}(F \circ F) \leq S_3 \wr S_3. \quad (5)$$

3 A degree-nine slice

Restrict the target to $(a, b, c) = (1, 2, s)$. Let t satisfy

$$2t^3 - 2t^2 + 2t - s = 0$$

and reconstruct $(x(t), y(t), z(t))$ using (4). An inner resolvent root r satisfies

$$2x(t)r^3 - y(t)r^2 + 2r - z(t) = 0.$$

Eliminating t and removing the vertical content $256s^7$ gives the primitive polynomial $P(r, s)$ below.

$$\begin{aligned}
P = & 128r^9s - 256r^8 + (2592s^2 - 3072s + 1408)r^7 \\
& + (-7452s^3 + 1248s^2 - 1584s - 576)r^6 \\
& + (-17496s^4 + 82944s^3 - 87936s^2 + 42240s - 5760)r^5 \\
& + (52488s^4 - 142884s^3 + 135144s^2 - 60144s + 8928)r^4 \\
& + (104976s^5 + 97200s^4 - 97056s^3 - 41472s^2 + 93824s - 26880)r^3 \\
& + (236196s^6 - 380538s^5 - 147744s^4 + 772080s^3 - 754944s^2 \\
& \quad + 333792s - 62208)r^2 \\
& + (236196s^6 - 734832s^5 + 1916784s^4 - 2658432s^3 + 2128320s^2 \\
& \quad - 919296s + 186624)r \\
& + 531441s^8 - 2204496s^7 + 5436882s^6 - 11805534s^5 + 17996446s^4 \\
& \quad - 17402304s^3 + 10364976s^2 - 3544992s + 557280.
\end{aligned}$$

Its roots generically parametrize the nine points in the slice fiber.

4 Three inertia elements

4.1 A 9-cycle at infinity

Write $v = 1/s$ and consider $v^8P(r, v^{-1})$. For the coefficients of $r^0, \dots, r^9$, their v -adic valuations are

$$0, 2, 2, 3, 4, 4, 5, 6, 8, 7.$$

Every intermediate point lies strictly above the line from $(0, 0)$ to $(9, 7)$. The lower Newton polygon is therefore one edge of slope $7/9$.

Set $v = w^9$ and $r = w^{-7}R$. The initial equation is

$$128R^9 + 531441 = 0.$$

Its roots are simple and lift uniquely to Puiseux branches. The local deck transformation $w \mapsto \zeta_9 w$ multiplies the leading term by ζ_9^{-7} . Since $\gcd(7, 9) = 1$, it permutes all nine branches in one orbit. Thus slice monodromy contains a 9-cycle α .

4.2 A single transposition

Exact discriminant factorization gives

$$\text{disc}_r(P) = 2^{38}q(s)^8A(s)B(s)^2, \quad q(s) = 27s^2 - 28s + 12, \quad (6)$$

where

$$\begin{aligned}
A(s) = & 1129718145924s^{12} - 6474958632657s^{11} + 21955119111630s^{10} \\
& - 60661410535386s^9 + 123515279853390s^8 - 191093865073182s^7 \\
& + 235490291194248s^6 - 188631015819696s^5 + 54561577345568s^4 \\
& + 41882989175328s^3 - 44714997684096s^2 + 15094447332352s \\
& - 1434819245056.
\end{aligned}$$

The exact certificates verify that A is squarefree and coprime to q , B , and the leading coefficient $128s$. At each root of A , the discriminant has valuation one while the degree remains nine. Local geometric inertia is thus a single transposition β . A permutation moving only two sheets cannot act nontrivially on the three blocks, so β is supported in one block.

The full degree-22 factor B is recorded in Appendix A and checked independently in SymPy and PARI/GP.

4.3 A transposition of the blocks

The outer cubic has discriminant

$$\text{disc}_t(2t^3 - 2t^2 + 2t - s) = -4(27s^2 - 28s + 12) = -4q(s).$$

The quadratic q is squarefree and coprime to A and B . A small loop about either root of q has outer monodromy a transposition. Total inertia contains an element γ whose image on the three blocks is that transposition.

5 The group argument

Lemma 4. *Let $H \leq S_3 \wr S_3$ in its natural action on three blocks of size three. Suppose H contains a 9-cycle α , a single transposition β , and an element γ inducing a transposition on the blocks. Then $H = S_3 \wr S_3$.*

Proof. The 9-cycle induces a 3-cycle on the blocks, and α^3 acts as a 3-cycle inside each block. The transposition β is supported in one block. Conjugating it by α^3 gives a distinct transposition in that block, so these elements generate the full S_3 supported there. Conjugation by powers of α gives the full S_3 independently in all three blocks. Hence the base subgroup S_3^3 lies in H .

On blocks, α supplies a 3-cycle and γ a transposition, so the image of H is the top S_3 . A subgroup containing the full base and surjecting onto the top is the full semidirect product. $\square$

The three inertia elements constructed above satisfy the lemma, so the slice monodromy is $S_3 \wr S_3$. Slice monodromy is a subgroup of generic monodromy; combining this with (5) proves Theorem 1.

6 Full-cycle inertia for all iterates

We prove Proposition 2. Put $v = 1/s$, start from $X_0 = (1, 2, s)$, and choose a compatible inverse tower $F(X_k) = X_{k-1}$. Write

$$x_k \sim c_x s^{-a_k}, \quad y_k \sim c_y s^{a_k}, \quad z_k \sim c_z s^{c_k},$$

where the constants are nonzero. Initially $a_0 = 0$ and $c_0 = 1$.

The next inverse parameter satisfies

$$2x_k t^3 - y_k t^2 + 2t - z_k = 0.$$

If $c_k > 5a_k$, the lower Newton edge balances the first and last terms, so

$$t_{k+1} \sim c_t s^{(a_k + c_k)/3}.$$

The reconstruction formulas then yield

$$a_{k+1} = \frac{c_k - 2a_k}{3}, \quad c_{k+1} = 2(c_k - a_k). \quad (7)$$

The hypothesis persists: for $\rho_k = a_k/c_k$, the interval $0 \leq \rho_k \leq 1/6$ is invariant under

$$\rho_{k+1} = \frac{1 - 2\rho_k}{6(1 - \rho_k)},$$

whose image on that interval is $[2/15, 1/6]$ after the first step. Thus no omitted term lies on the lower edge and no leading coefficient cancels.

Set $A_k = 3^k a_k$ and $C_k = 3^k c_k$. Equation (7) becomes

$$A_{k+1} = C_k - 2A_k, \quad C_{k+1} = 6(C_k - A_k), \quad A_0 = 0, \quad C_0 = 1.$$

The exponent of t_m is

$$\frac{E_m}{3^m}, \quad E_m = A_{m-1} + C_{m-1}.$$

For $m \geq 2$, C_{m-1} is divisible by three and $A_{m-1} \equiv 1 \pmod{3}$. Hence $3 \nmid E_m$. The first exponents are

$$\frac{1}{3}, \quad \frac{7}{9}, \quad \frac{34}{27}, \quad \frac{178}{81}, \dots$$

At level m , the local inverse field has ramification index at least 3^m ; the global degree is at most 3^m , so equality holds and the local extension is totally ramified. Finite extensions of $\mathbb{C}((v))$ are tame Puiseux extensions, so inertia acts as a single 3^m -cycle. This proves the proposition.

As an independent check, nested PARI resultants for $m = 3$ give a degree-27 eliminant with one lower Newton edge from $(0, 0)$ to $(27, 34)$.

7 Independent arithmetic cross-check

At $s = -3$, the primitive specialization is

$$\begin{aligned} f(r) = & -384r^9 - 256r^8 + 33952r^7 + 216612r^6 - 4580568r^5 \\ & + 9515052r^4 - 15697056r^3 + 223986114r^2 + 599887620r \\ & + 17172300267. \end{aligned}$$

Dependency-free modular arithmetic verifies squarefree factorizations with irreducible-degree patterns

p	degrees
13	(9)
61	(2, 1, 1, 1, 1, 1, 1, 1)
19	(2, 2, 2, 2, 1).

These give Frobenius cycle types (9), $(2, 1^7)$, and $(2^4, 1)$. The same group lemma forces the arithmetic group to be the full wreath product. GAP independently enumerates all subgroups of the wreath product and finds no proper survivor. PARI/GP 2.17.4 identifies five unrelated rational fibers as transitive group 9T31 of order 1296.

This is corroboration only; the geometric proof does not infer geometric monodromy from an arithmetic specialization.

8 Verification and priority scope

The artifact contains four independent checks:

File	Scope
<code>verify_symbolic.py</code>	Jacobian, collision, inverse identities, resultant, Newton data, discriminant, squarefreeness, coprimality
<code>verify_modular.py</code>	dependency-free polynomial arithmetic and Rabin irreducibility certificates
<code>verify_pari.gp</code>	independent discriminant, coprimality, finite-field, and arithmetic Galois checks
<code>verify_group.g</code>	exhaustive subgroup check inside $S_3 \wr S_3$

The optional `verify_level3_newton.py` checks the degree-27 edge with SymPy-generated equations and PARI resultants.

The closest public monodromy study states only $\text{Mon}(F_0 \circ F_0) \leq S_3 \wr S_3$ and calls its exact group a natural next computation [2]. A separate four-variable construction already claims a different imprimitive degree-eight group of order 192 [3]. We therefore do not claim the first imprimitive or first non-symmetric Keller monodromy. The claimed contribution is the exact canonical self-composition group, the realization $(9, S_3 \wr S_3)$, and the all-iterate full-cycle statement. A time-stamped 23-repository audit accompanies the artifact.

The stronger arboreal assertion that every iterate has the full iterated wreath product remains open: the present argument does not yet produce a new single transposition at every level.

A The degree-22 square factor

$$\begin{aligned}
B(s) = & 1423119505038213888s^{22} - 155058052818404824443s^{21} + 1413754646027656083066s^{20} \\
& - 8145658955220494785812s^{19} + 33677018812011807334224s^{18} - 108385682498649366622416s^{17} \\
& + 280859820926917245978240s^{16} - 598631883156564470992728s^{15} + 1062288637590844067815584s^{14} \\
& - 1579910926841521168581900s^{13} + 1974730823883289142626824s^{12} - 2073348574060015592229024s^{11} \\
& + 1822485349587628391880960s^{10} - 1333523184375693801555072s^9 + 806144943610022071172864s^8 \\
& - 398941760796329492797440s^7 + 159756878618286560778240s^6 - 50952869532774134959104s^5 \\
& + 12636860641575849510912s^4 - 2344566242453066735616s^3 + 304677812001210531840s^2 \\
& - 24551525690267664384s + 926711257485017088.
\end{aligned}$$

References

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